

Animal behavior analysis in python

Day1: Fundamental concept and introduction of behavior analysis

Miao Wang

Wang.miao@campus.lmu.de

Outline

- What is behavior
- Features of behavior
- Why it is import to analyze behavior?
- How to analyze behavior
 - Brief overview
 - Dataset introduction

1. What is behavior?

“Behavior is the total movements made by the intact animal”

--- Nikolaas Tinbergen



What will it mean to say we 'understand' the mechanism of behaviors?

	Genetic	Computer	Animal behavior
<i>Deeper level</i>	DNA ..	Transistor, integrated circuit	Neural circuit, synapses..
<i>General explanations</i>	Major principles of genetics	Software explanations	

**Hierarchical organisation:
a candidate principle for ethology**

RICHARD DAWKINS

2. Features of behavior --- hierarchical

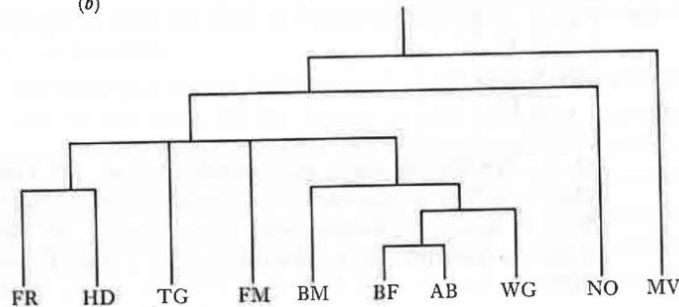
Data on grooming behaviour in blowflie

(a)

	Following									
	FR	TG	HD	FM	NO	MV	BM	AB	WG	BF
FR		77	709	129	496	5				
TG	82				2					
HD	730	4			18					
FM	151		1		13					
NO	445	3	42	36		418	26	133	495	414
MV	7				412					1
BM					13					76
AB					93				3	195
WG					223	1	1	46		384
BF					741		62	92	156	

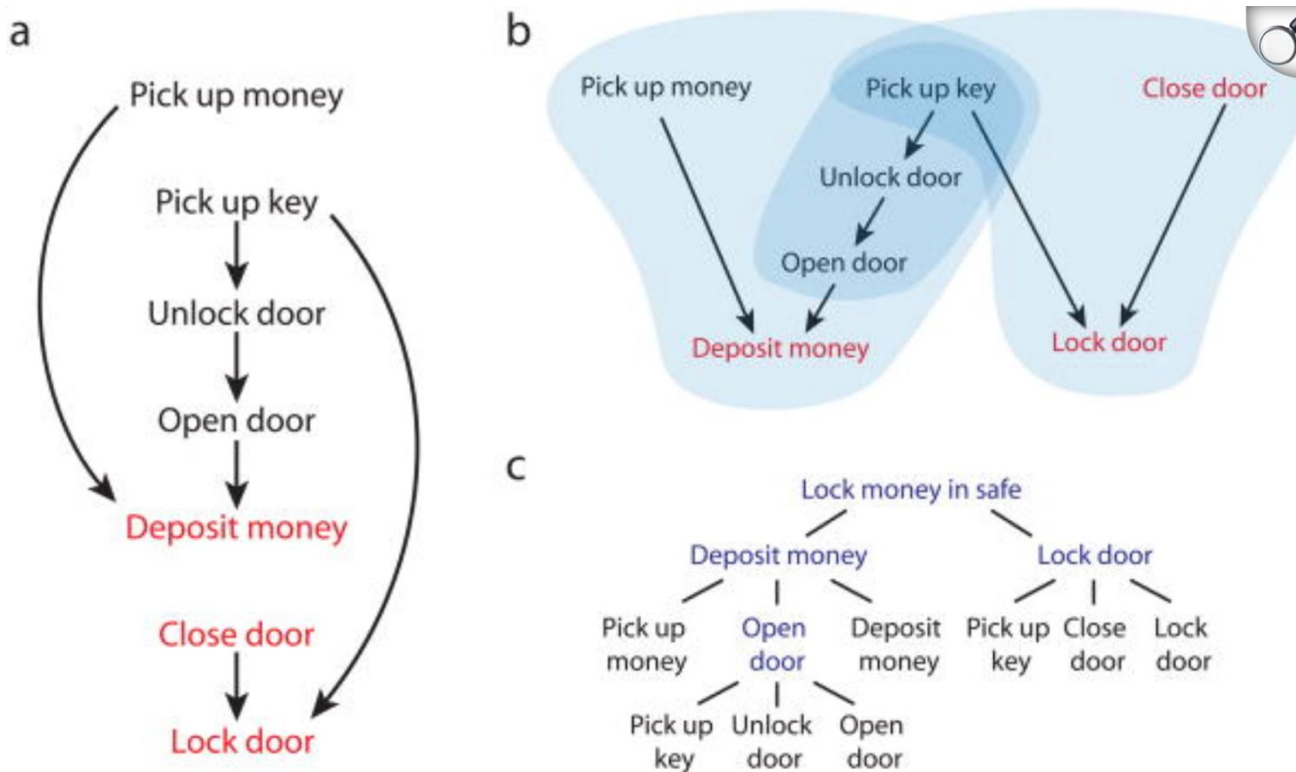
Behaviour patterns tend to occur clustered in time, the clusters constituting functionally related groups.

(b)



Dawkins

Two Kinds of Hierarchical Structure in human behavior

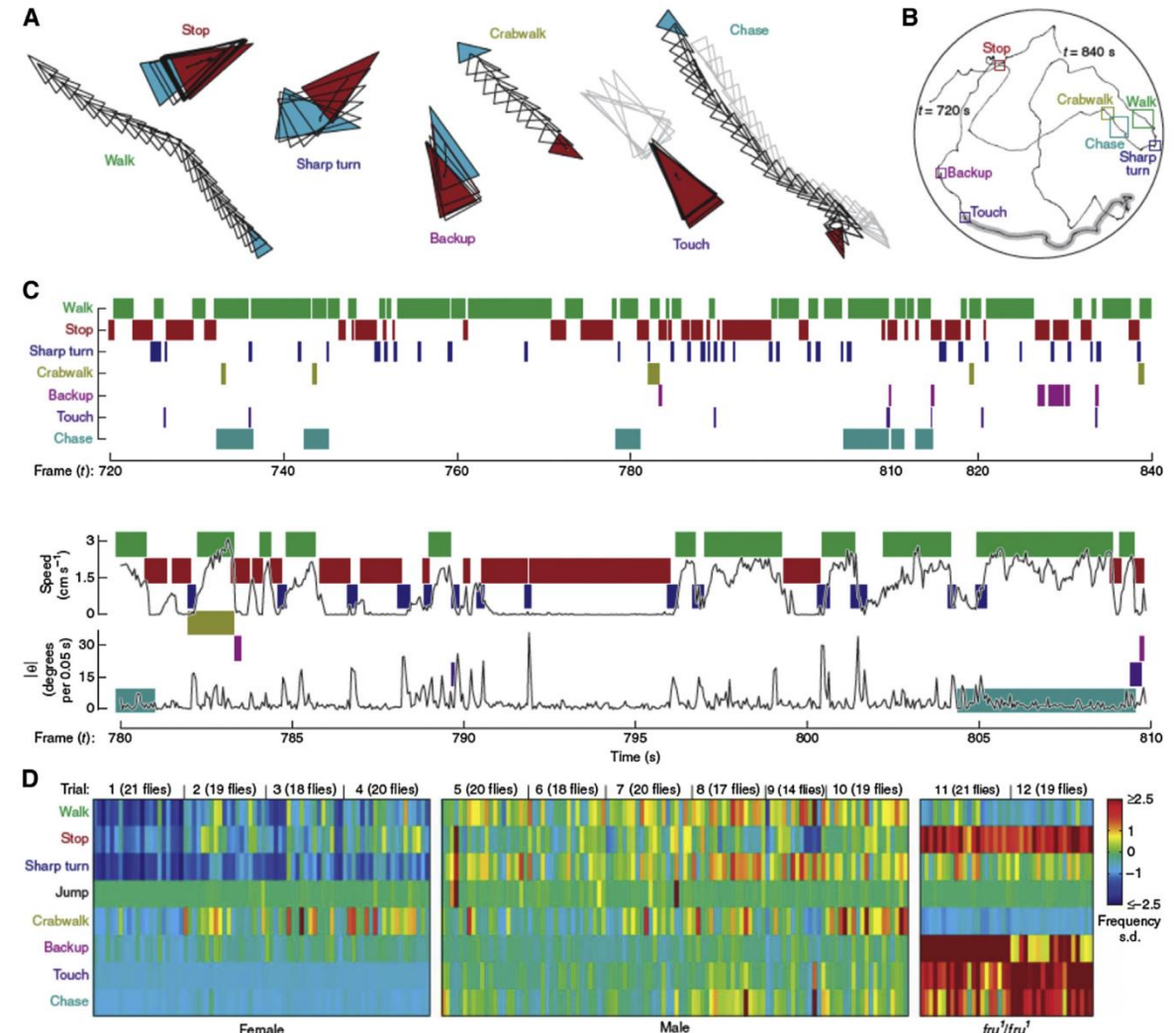


- *instrumental* structure(**Nested Decisions**): in the sense that action sequences bring about valued or desired outcomes
- *correlational* structure(**Temporal Scaling**) : by statistical co-occurrence

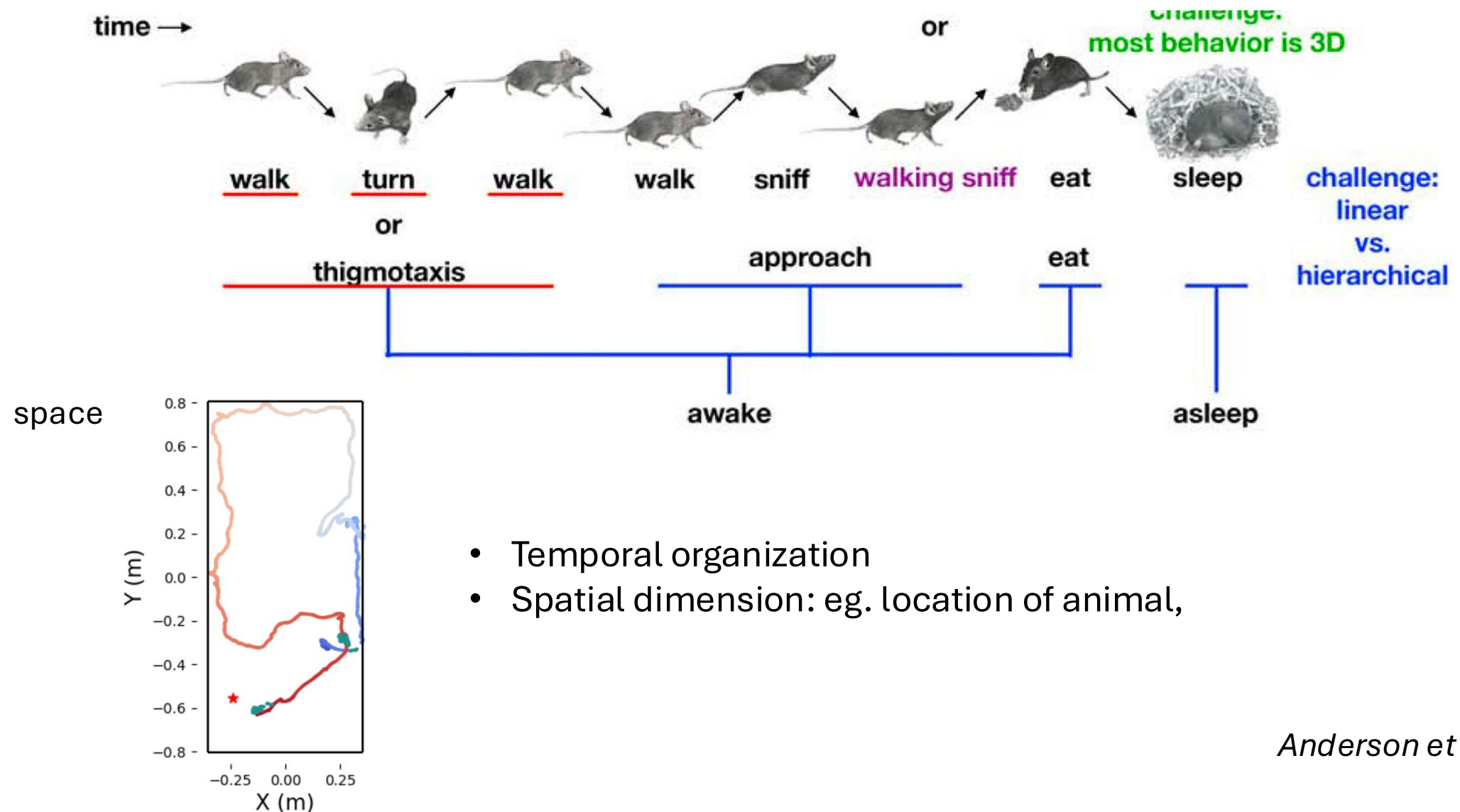
2. Features of behavior : structure/language for behavior

- Moveme: The simplest meaningful pattern associated with a behavior. (such as a turn, a step, or a wing extension)
- Action: A combination of movemes that always occurs in the same stereotypical sequence and that is also described by a verb. (“walk”)
- Activity: A species-characteristic concatenation of actions and movemes. (Examples of activities include courtship, aggression, nest-building, parenting, etc.)

Anderson et al. 2014

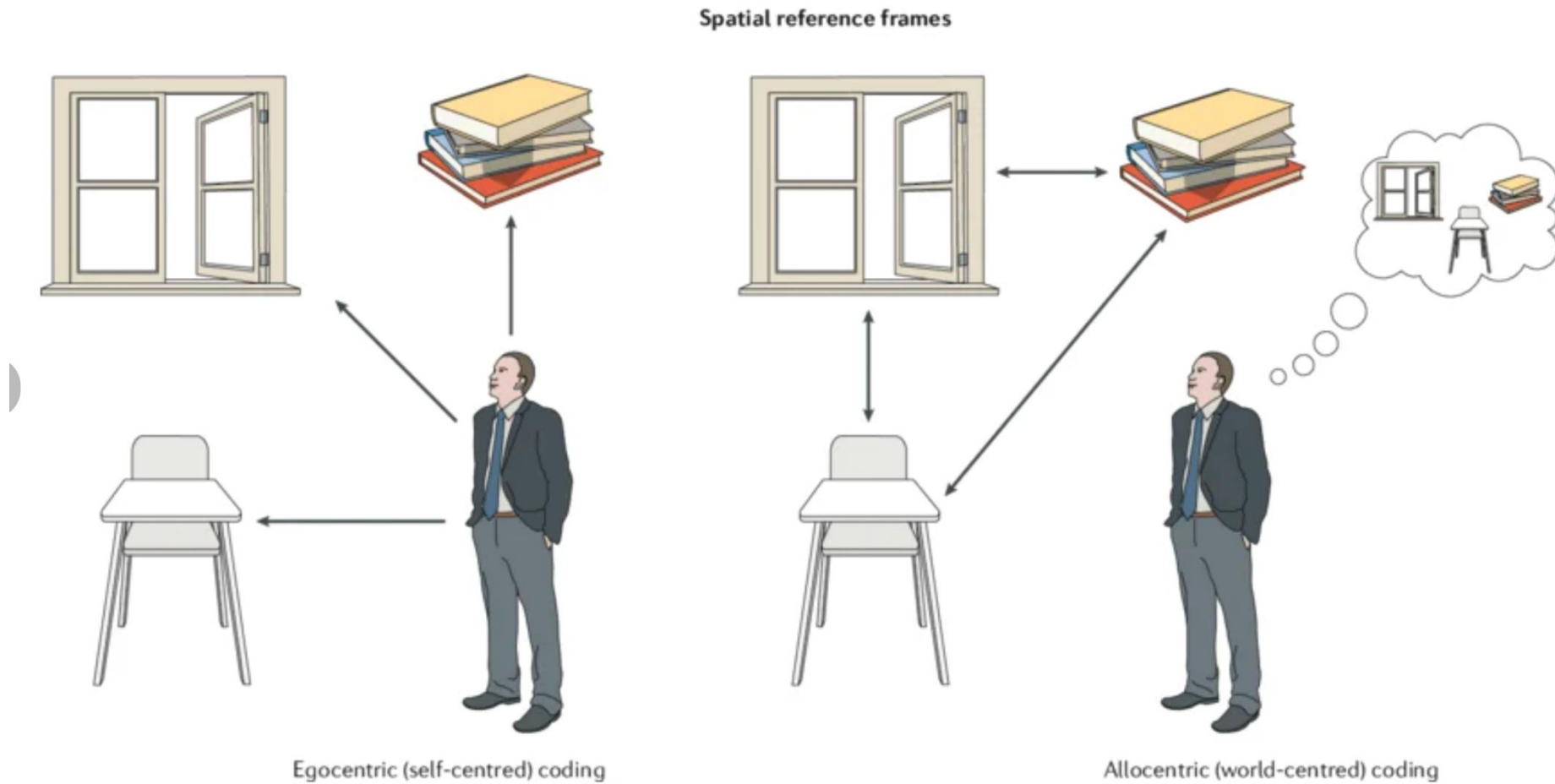


2. Features of behavior : temporal and spatial dimension

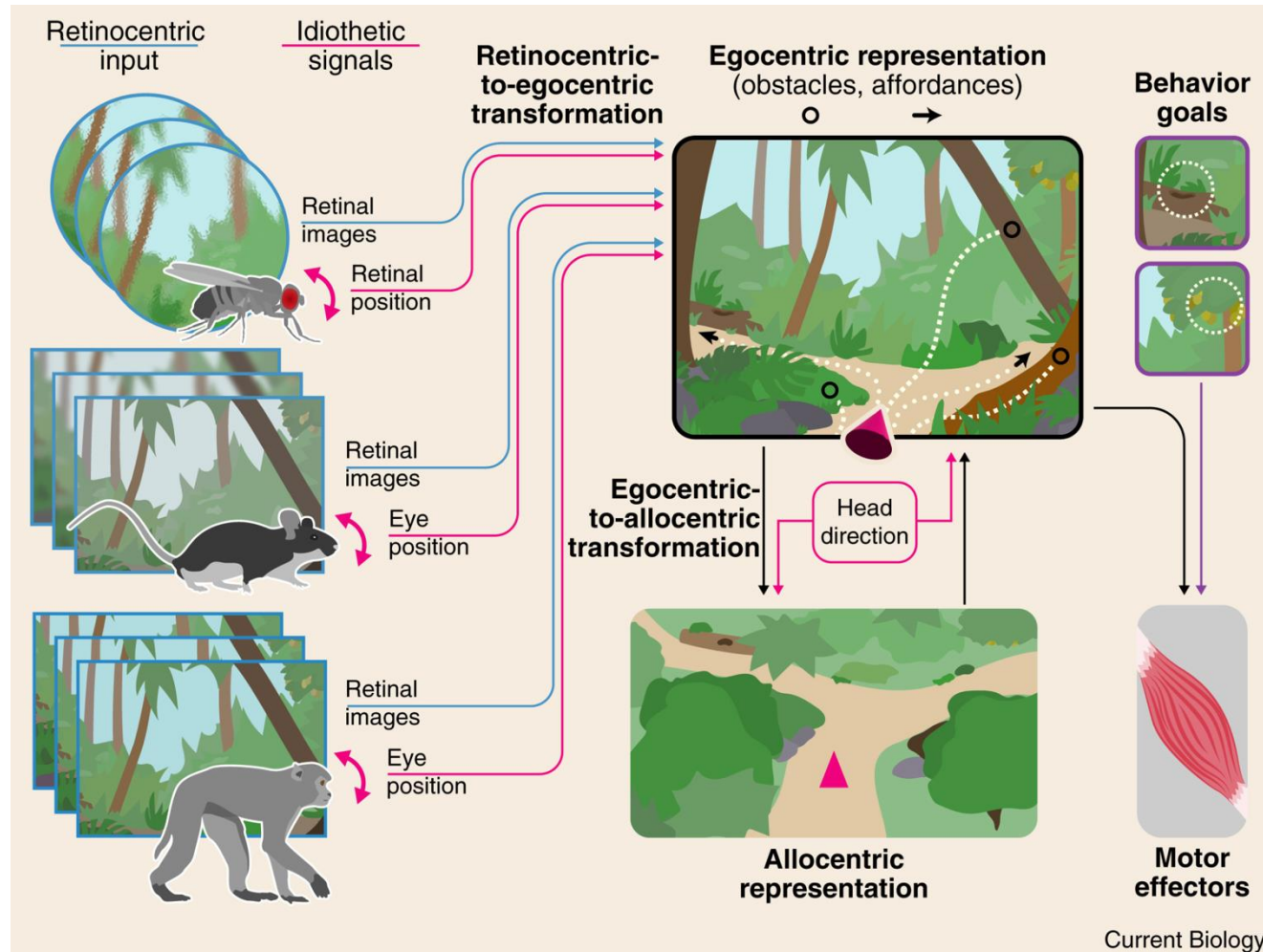


Anderson et al. 2020

2. Features of behavior : Different reference frames

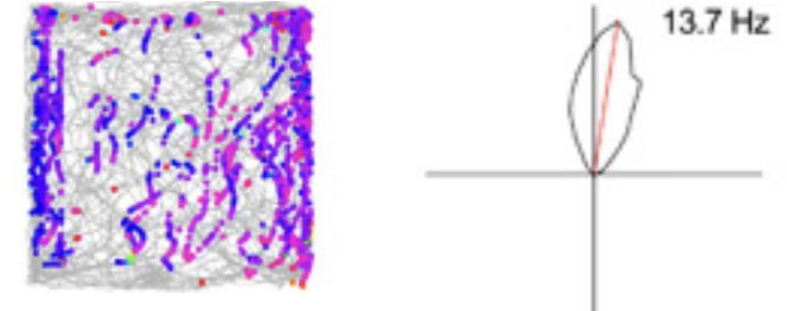


Reference frame transformations underlying the use of sensory input to guide motor output

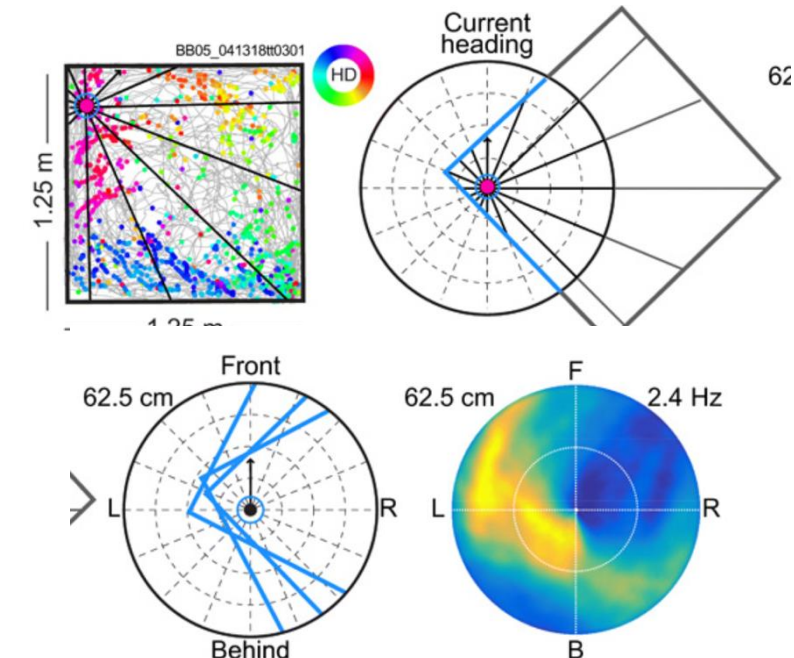


Martins et al. 2024

Allocentric: Head direction cells



Egocentric boundary vector cells

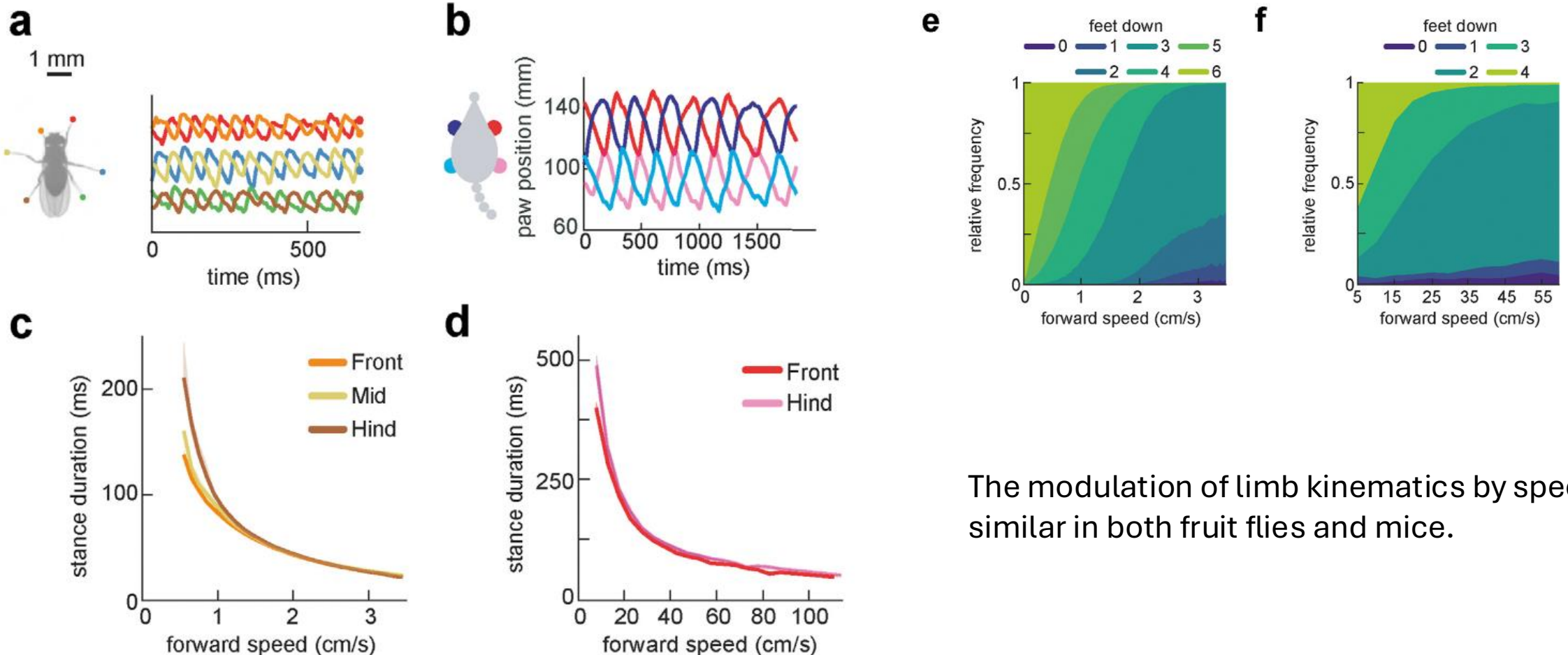


Alexander et al. 2020

2. Behavior feature: can be similar across animals

Parallel locomotor control strategies in mice and flies

Ana I. Gonçalves^{1,a}, Jacob A. Zavatone-Veth^{2,3,a},
Megan R. Carey^{1,b} and Damon A. Clark^{4,5,6,b}

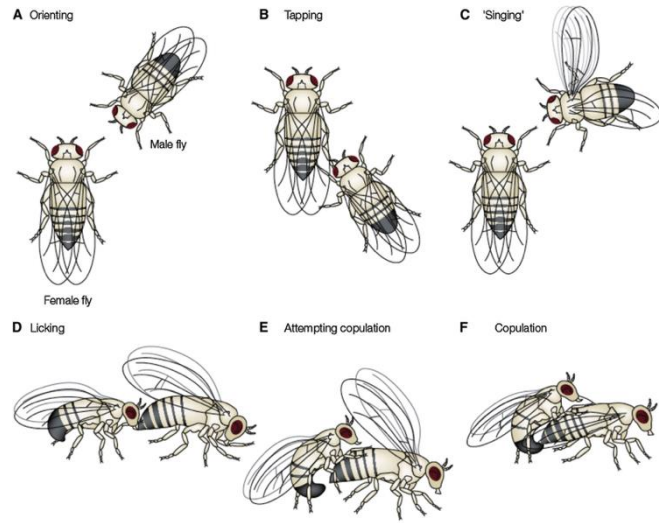


The modulation of limb kinematics by speed is similar in both fruit flies and mice.

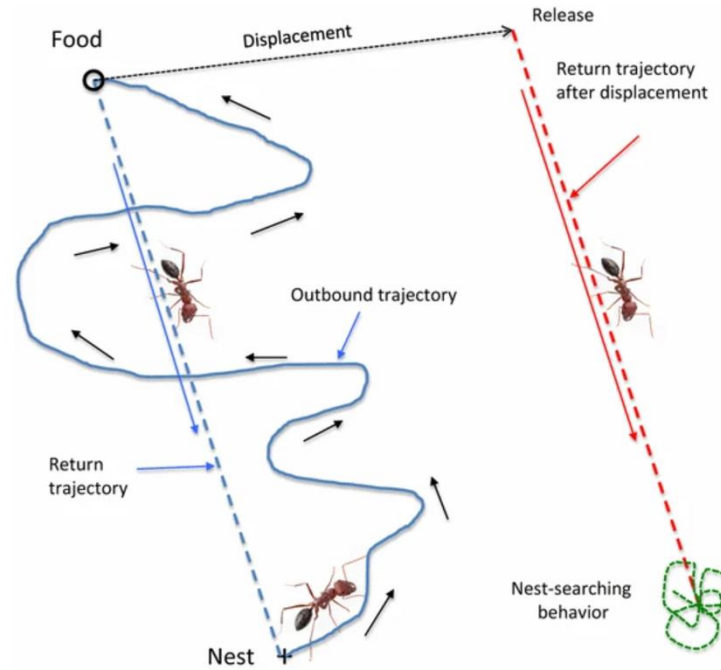
Midterm summary

- Feature of behavior
 - Hierarchical
 - Spatial and temporal dimension
 - Egocentric and allocentric reference frame
 - Share some common features across animals

3. Why study behavior?



Fruit fly courtship
Anderson et al. 2020



Ants path-integration

Mandyam V. Srinivasan 2015



Geese flying in v-formation



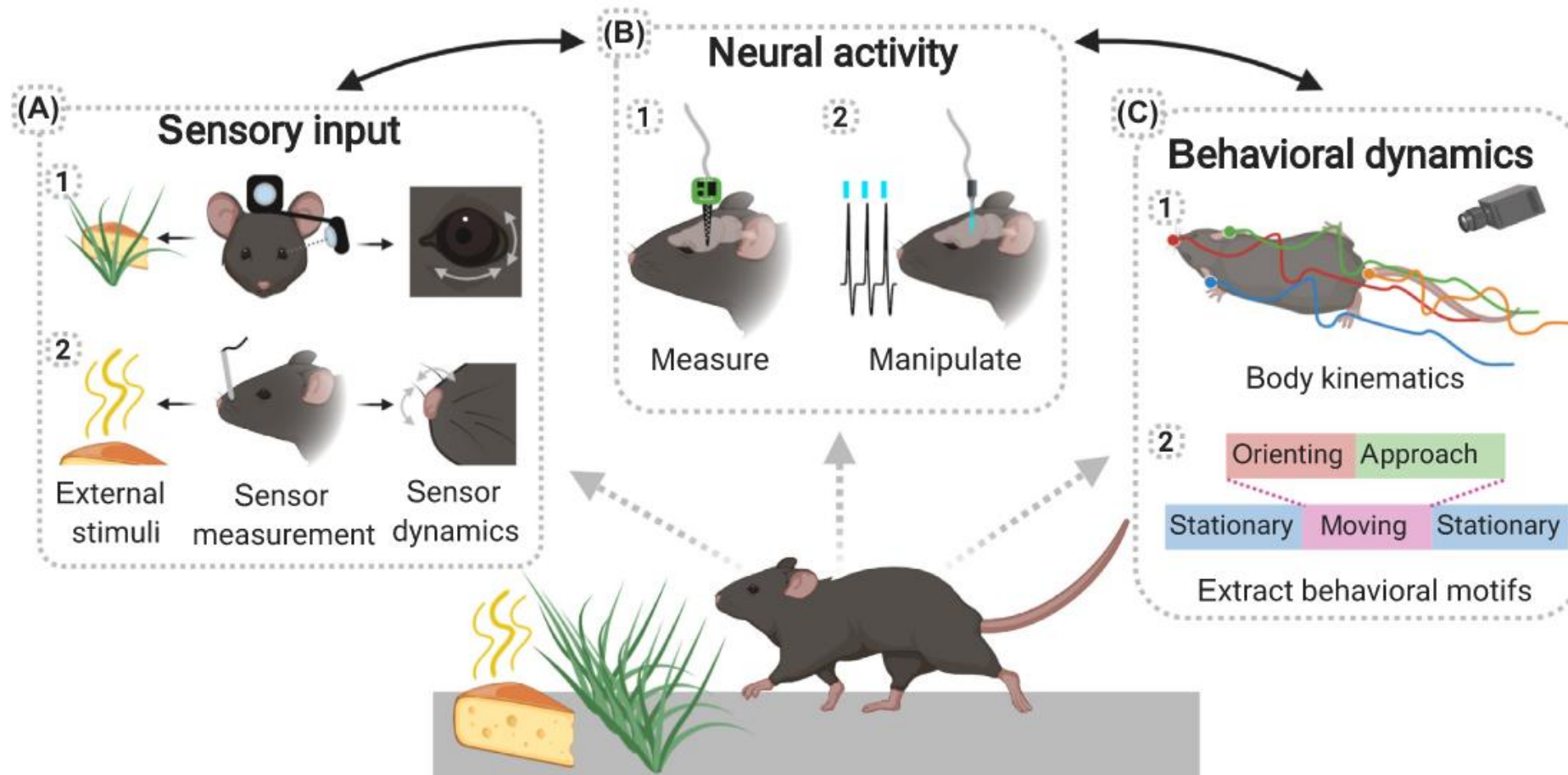
Penguins huddling in the Antarctic

wikipedia

Ecological meaning: to understand how animal response to the ecological pressures/environment and survive.

3. Why study behavior?

Connection to neuroscience

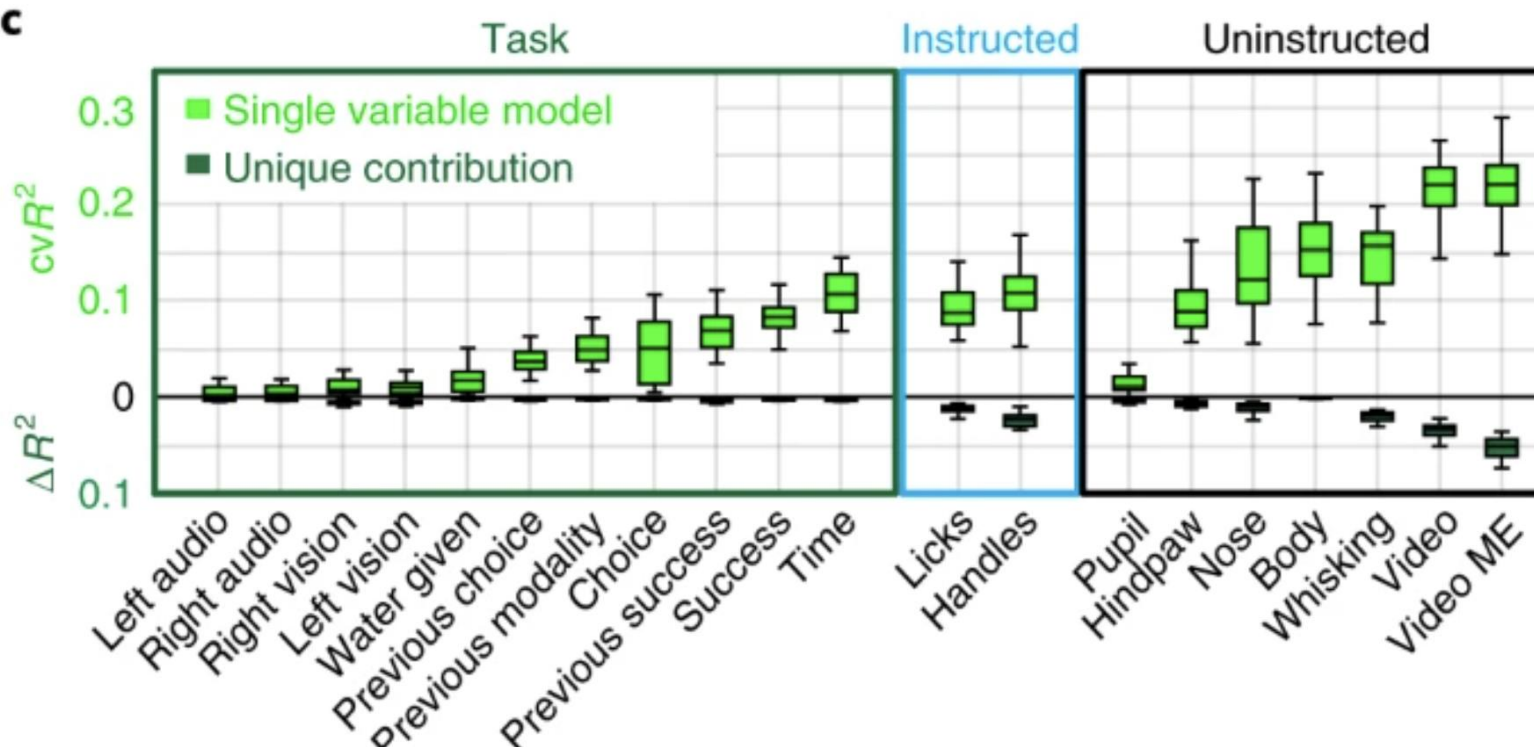
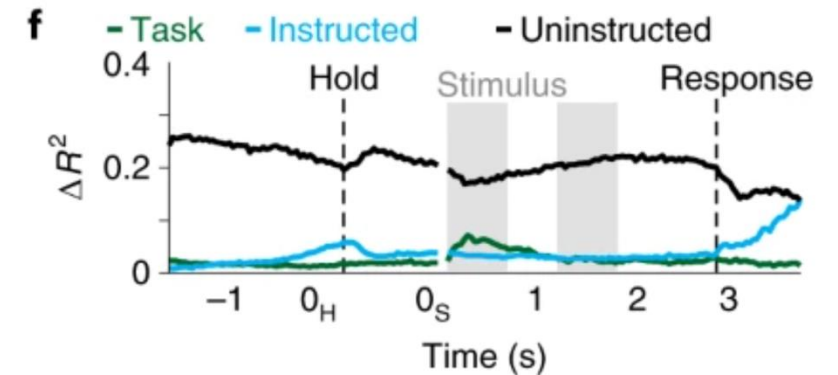
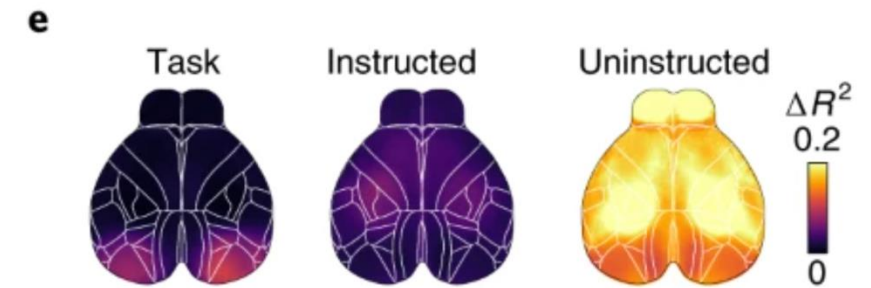
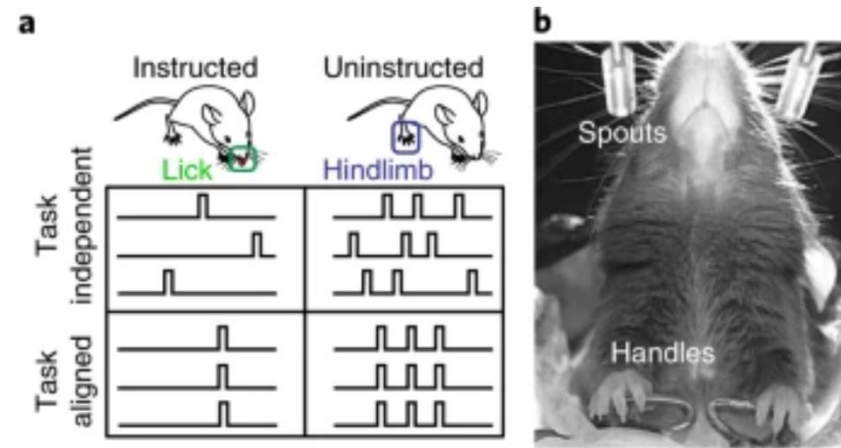


Behaviors explain(drive) the neural activities

Article | Published: 24 September 2019

Single-trial neural dynamics are dominated by richly varied movements

[Simon Musall](#), [Matthew T. Kaufman](#), [Ashley L. Juavinett](#), [Steven Gluf](#) & [Anne K. Churchland](#) 



Simon Musall et al. 2019

Infer the presence of internal states from observable behavior

Attention, arousal

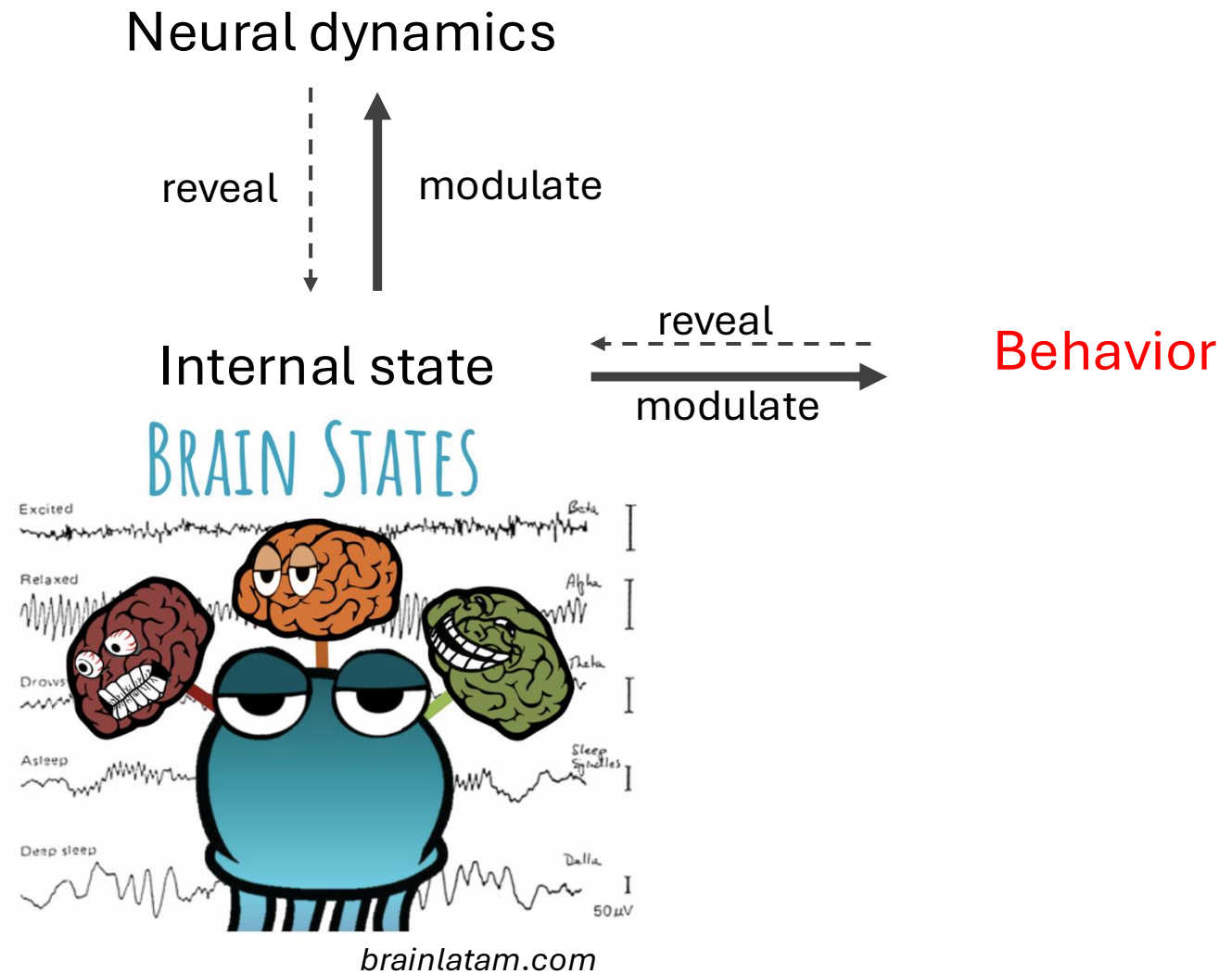
Enhancing sensory processing,
cognitive control ...

Sleep vs Wake

memory consolidation, cellular
repair, VS sensory processing,
learning, and adaptive behavior

Emotions: fear, ...

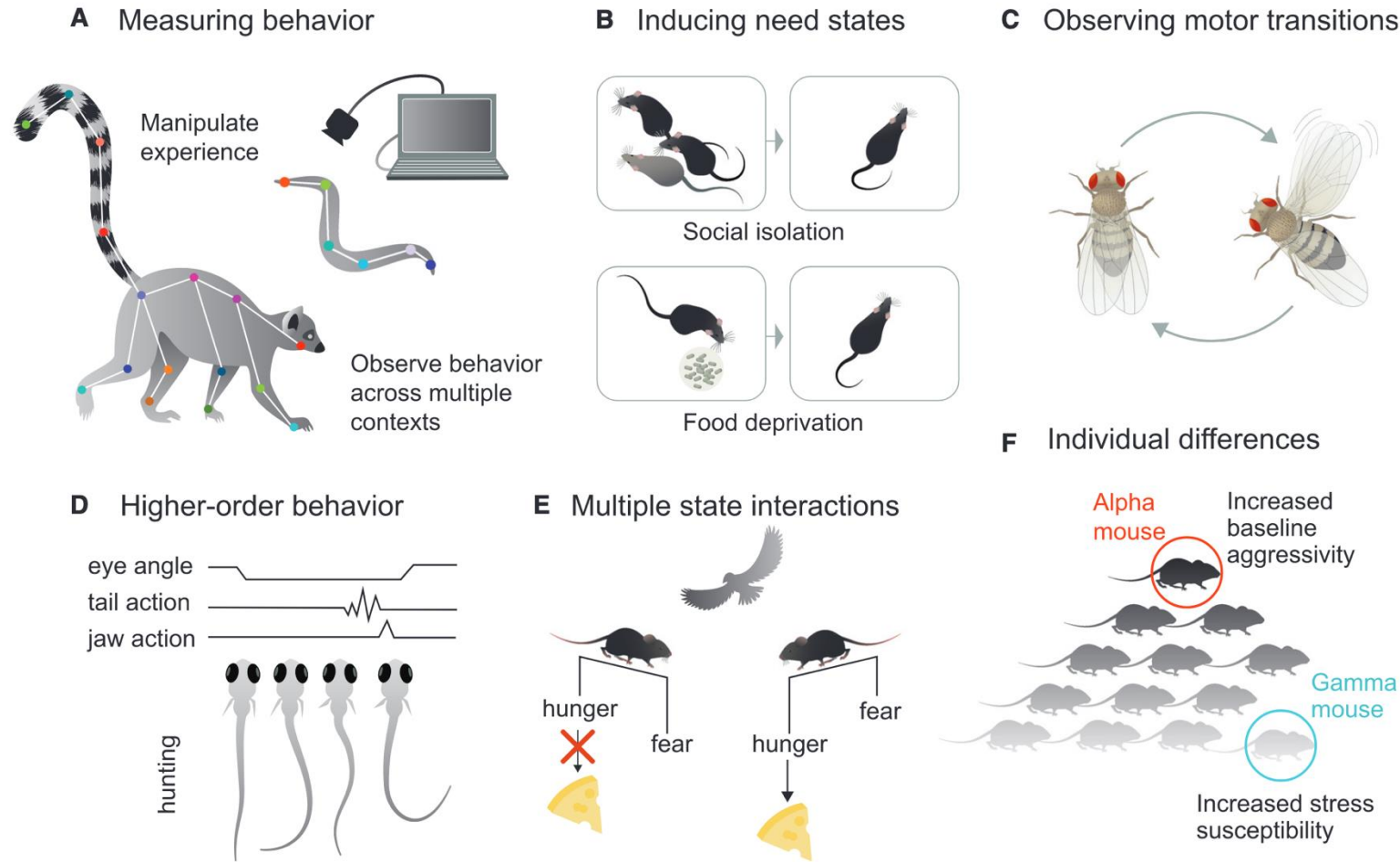
Crucial roles in survival, decision-making,
and social interactions



- Internal states shape perception, cognition, and action.
- Infer the presence of internal states from observable behavior

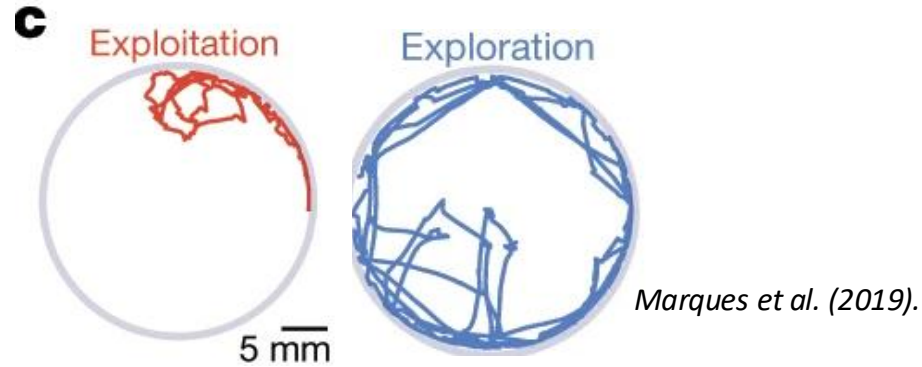
3. Why study behavior?

Infer the presence of internal states from observable behavior

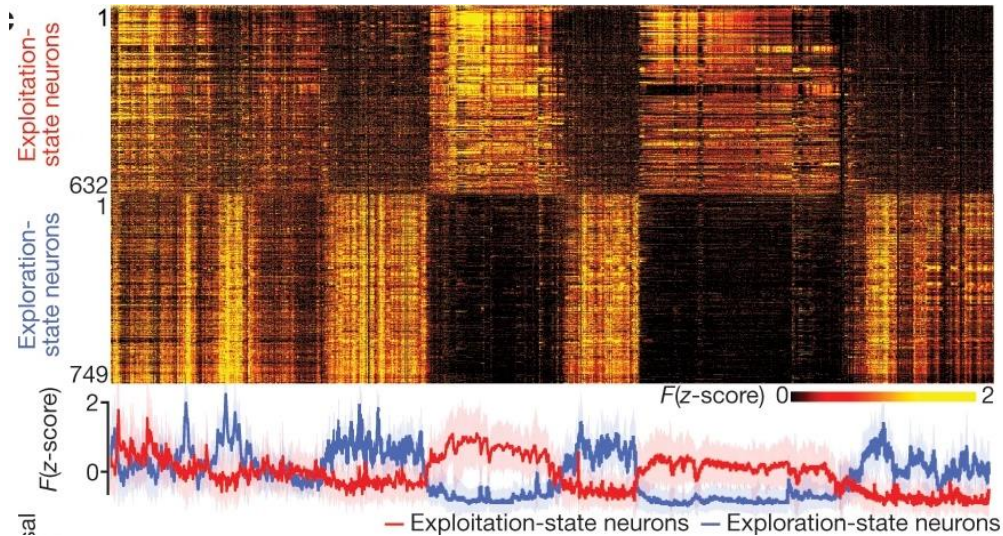


Classical states inferred based on behavior

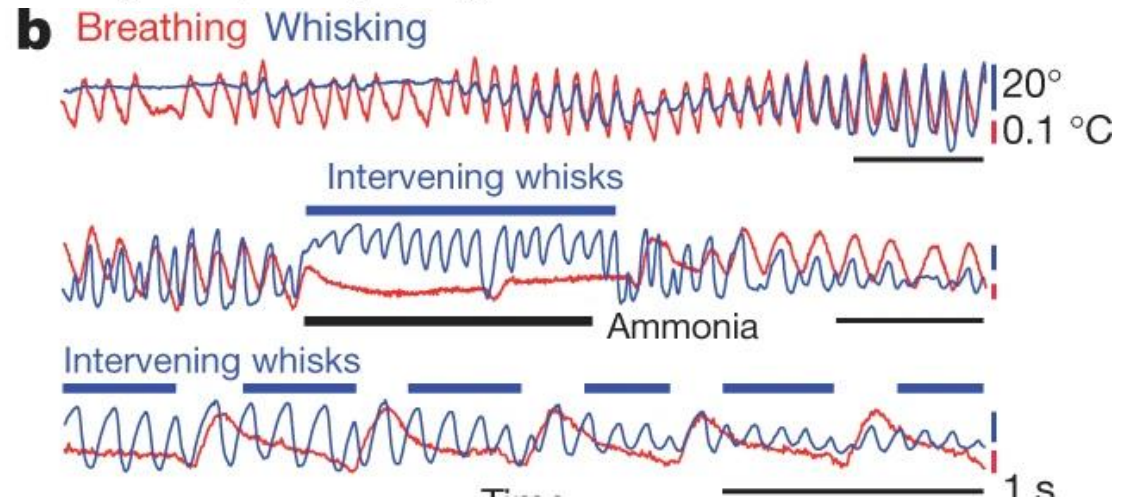
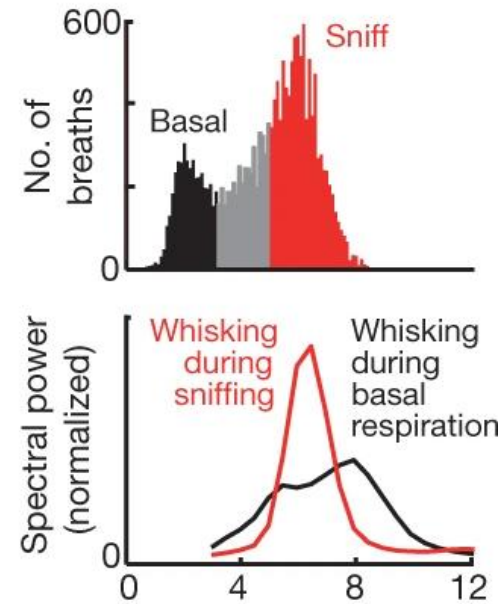
Zebrafish: use trajectory to define exploration vs exploitation



Zebrafish: brainwide activities

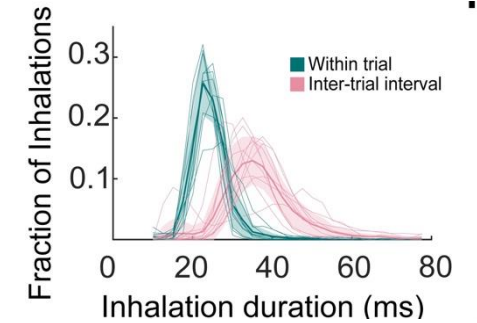


Marques et al. (2019).



Moore et al. (2013).

Rodents: sniffing, whisking to infer active sensing state



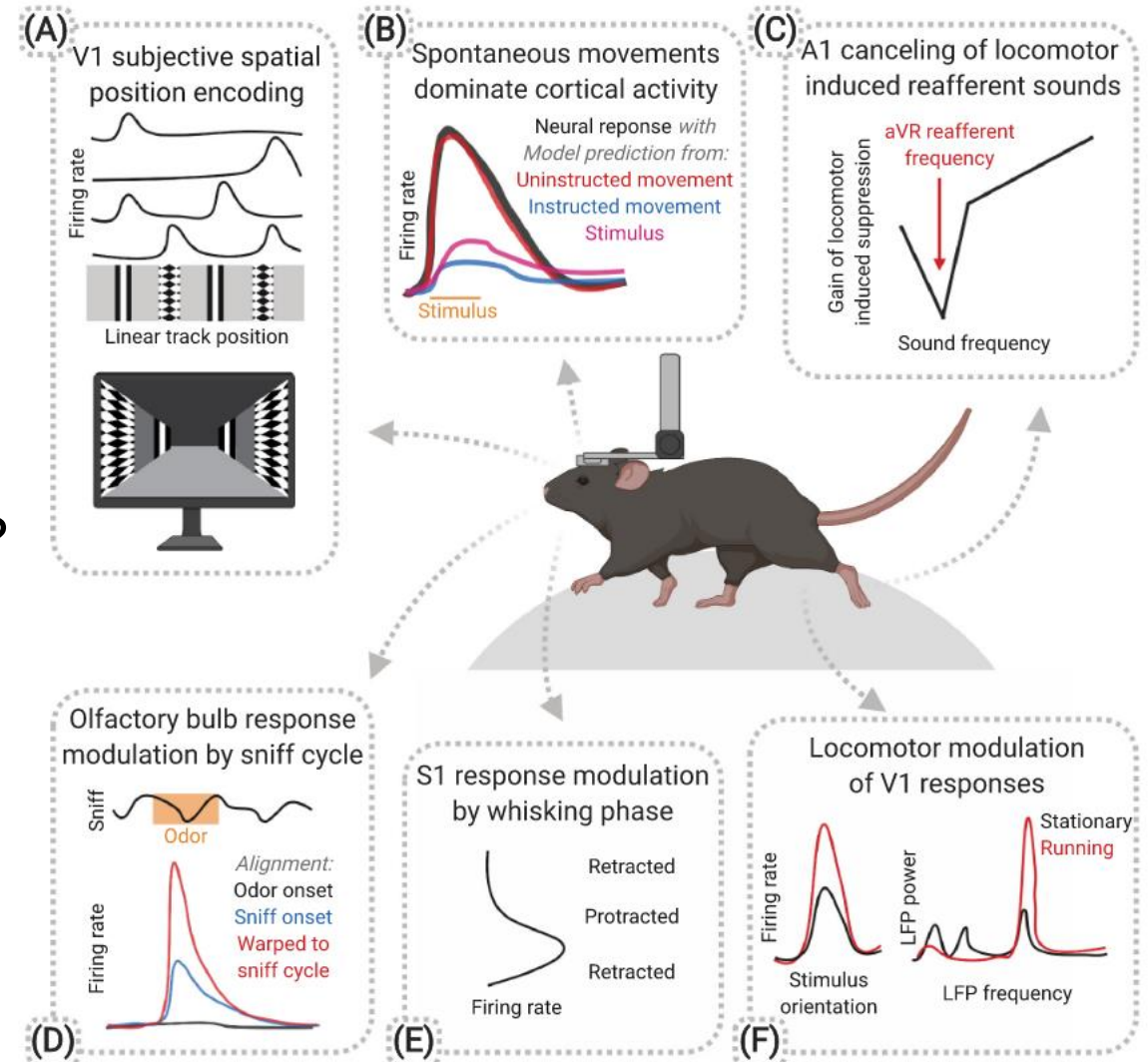
Findley et al. (2021).

Midterm summary

- Feature of behavior
 - Hierarchical
 - Spatial and temporal dimension
 - Egocentric and allocentric reference frame
 - Share some common features across animals
- Why analyze behavior
 - Behavior as an output: ecological meaning
 - Behavior that connect to neural activities
 - Behavior help to explain neural activities
 - Behavior as an window to infer the internal state

It is important to have a goal/question when analyze behavior.

- From ecology perspective
 - Why are they doing the v-shape?
 - How is the behavior organized?
- From neuroscience perspective:
 - How are the neuron related to the behavior?
 - What is the neurons drive this action?
 - Action – internal state – modulation
- Different features
 - Whisking, speed, pupils
 - Interacting with the environment (spatial dimension)

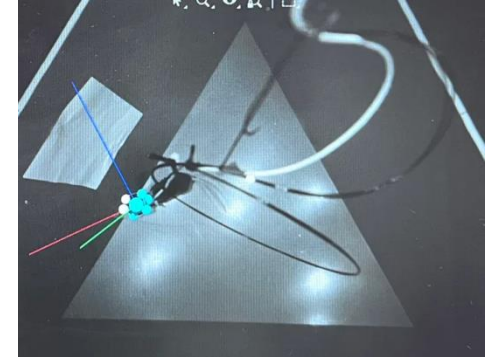


4. How to analyze behavior?

----Data availability

- Mark
- Moveme: The simplest meaningful pattern associated with a behavior. (such as a turn, a step, or a wing extension)
 - Action: A combination of movemes that always occurs in the same stereotypical sequence and that is also described by a

Video based



From frame based trajectory to explainable/interpretable activities.

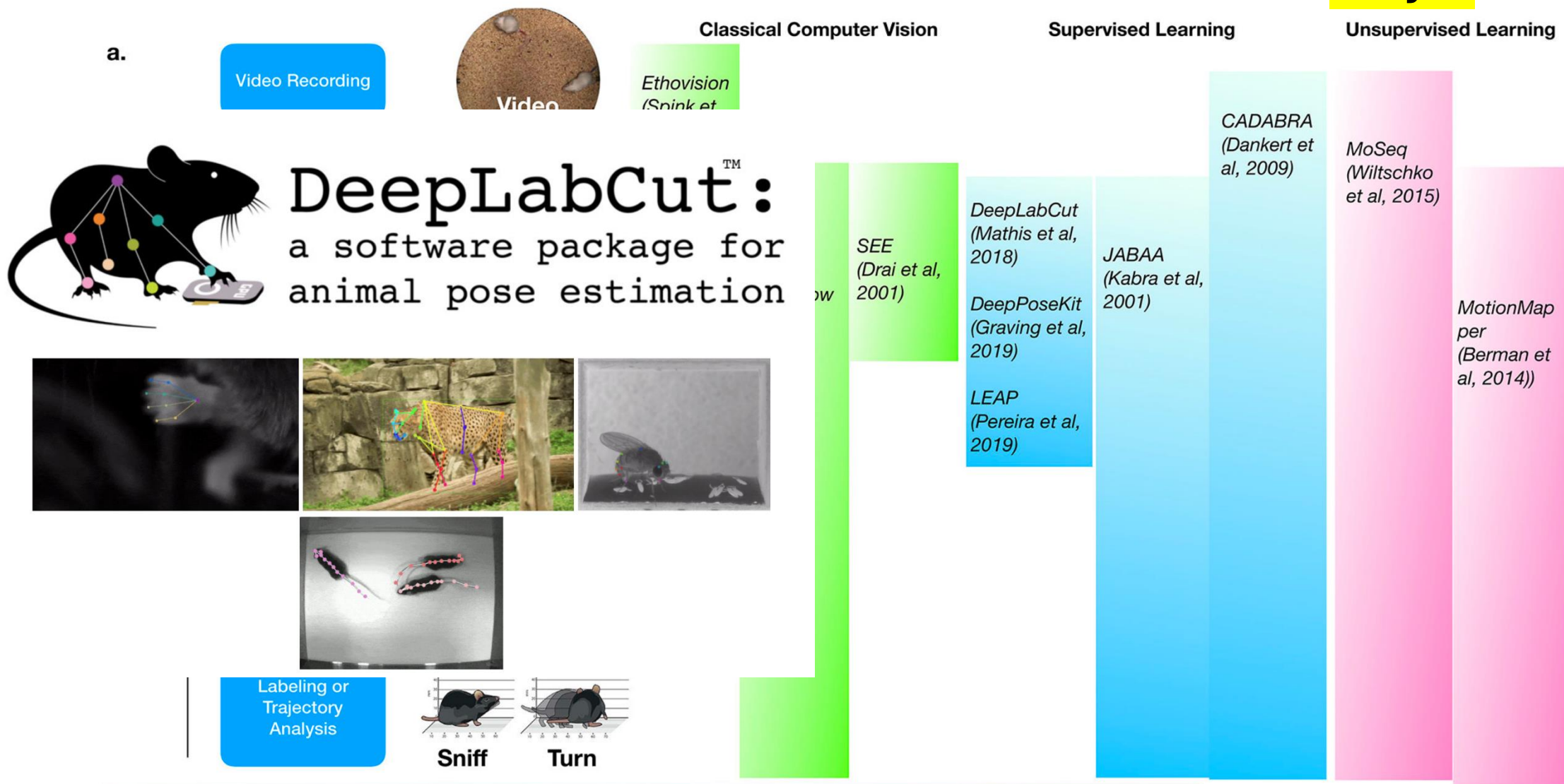
- Activity: A species-characteristic concatenation of actions and movemes. (Examples of activities include courtship, aggression, nest-building, parenting, etc.)

Anderson et al. 2014

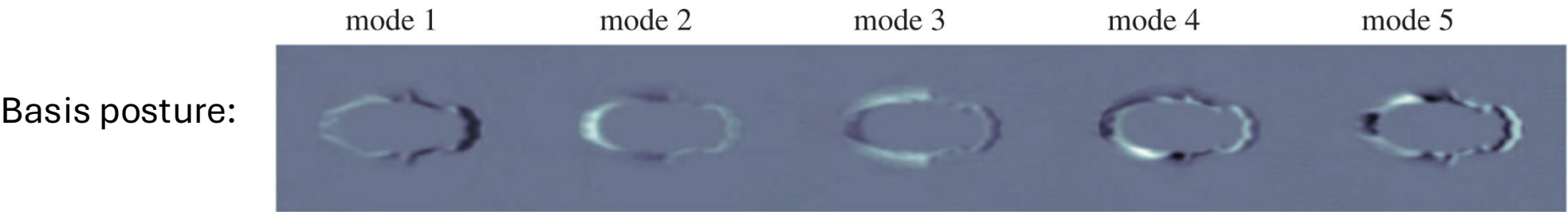
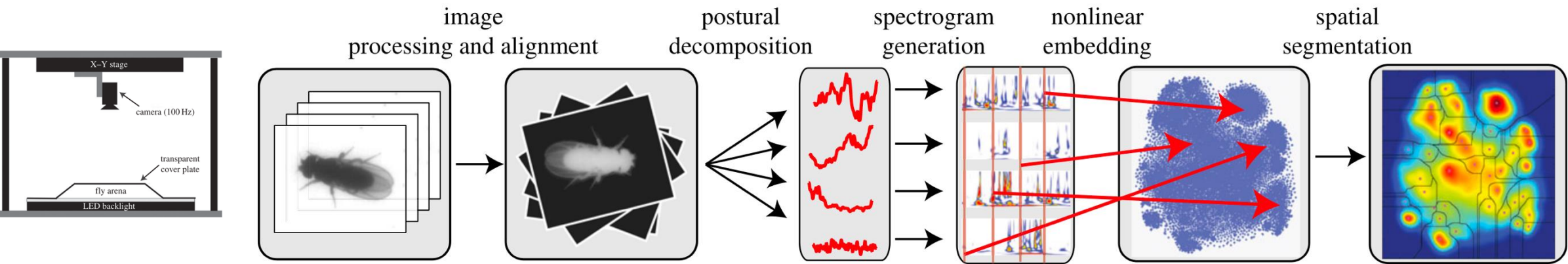
Day 2

Available method

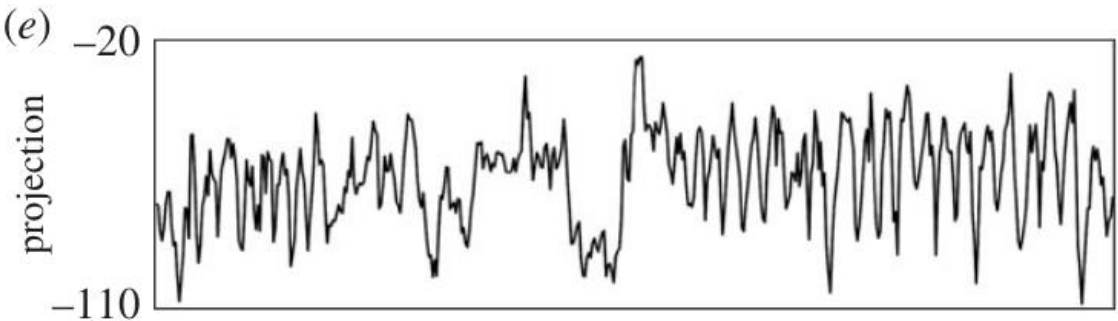
Day 4



Example 1: Mapping the stereotyped behaviour of freely moving fruit flies

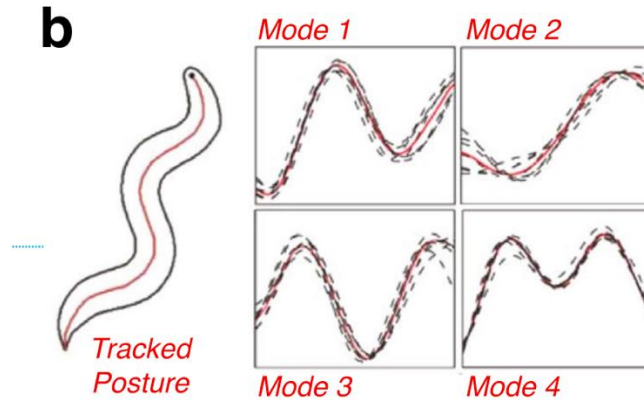
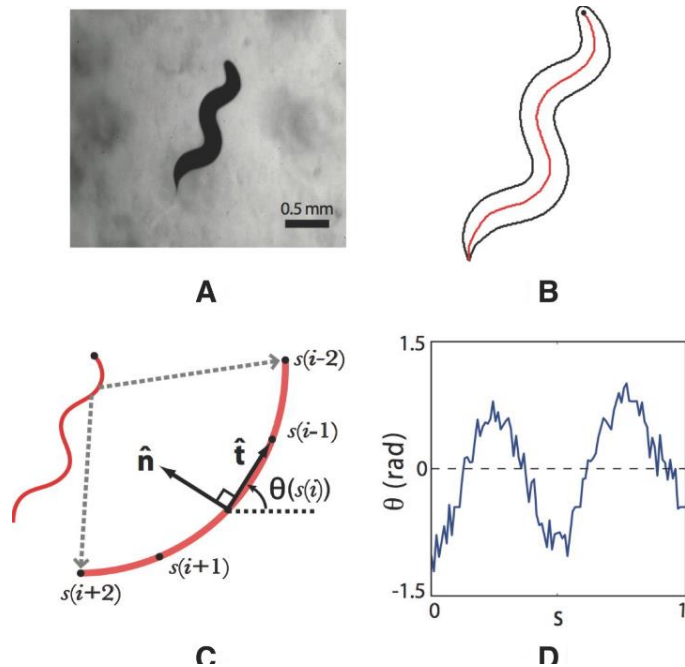


Contribution of certain basic mode to behavior



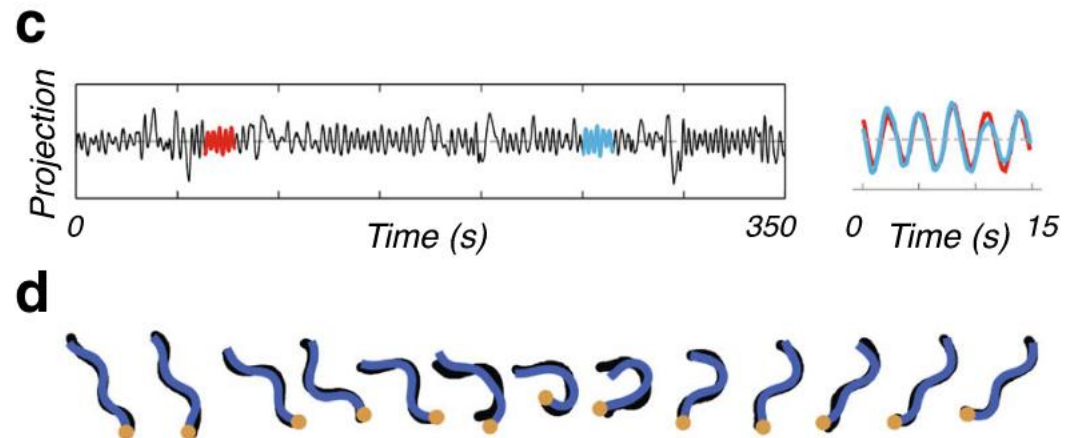
Berman et al. 2014

Example 2. C. elegans: finding basic mode on tracked posture



Reducing the Dimension of Tracked Postures

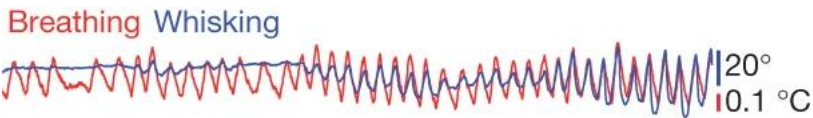
$$\theta(s) \approx \sum_{\mu=1}^K a_{\mu} u_{\mu}(s),$$



Example3 Behavioral features associated with exploratory states

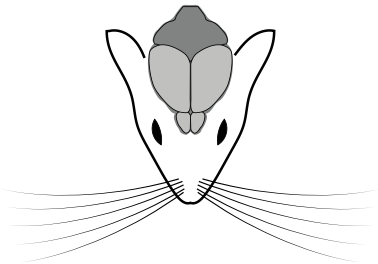
Moore et al. (2013).

Sniffing
(breath >5Hz)



Exploration

Whisking



Active sensing

Indicator for exploratory behavior

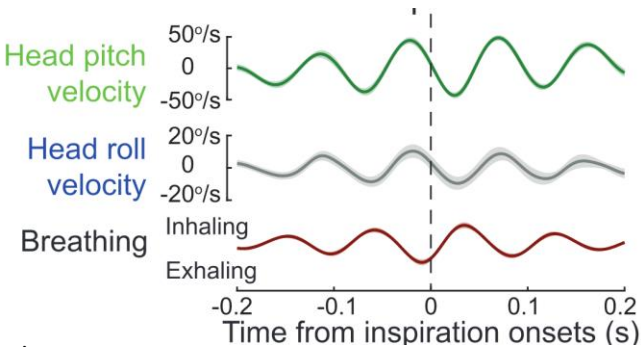
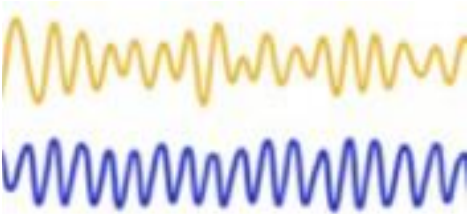
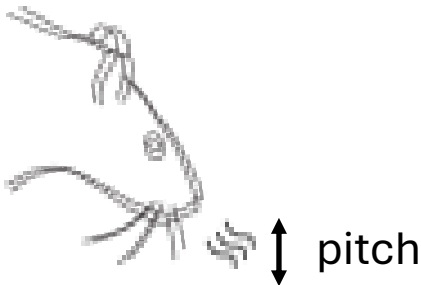
High breathing frequency
~ pitch frequency
~ whisking frequency

Day3

Head movement

Head pitch
(6-12 Hz)

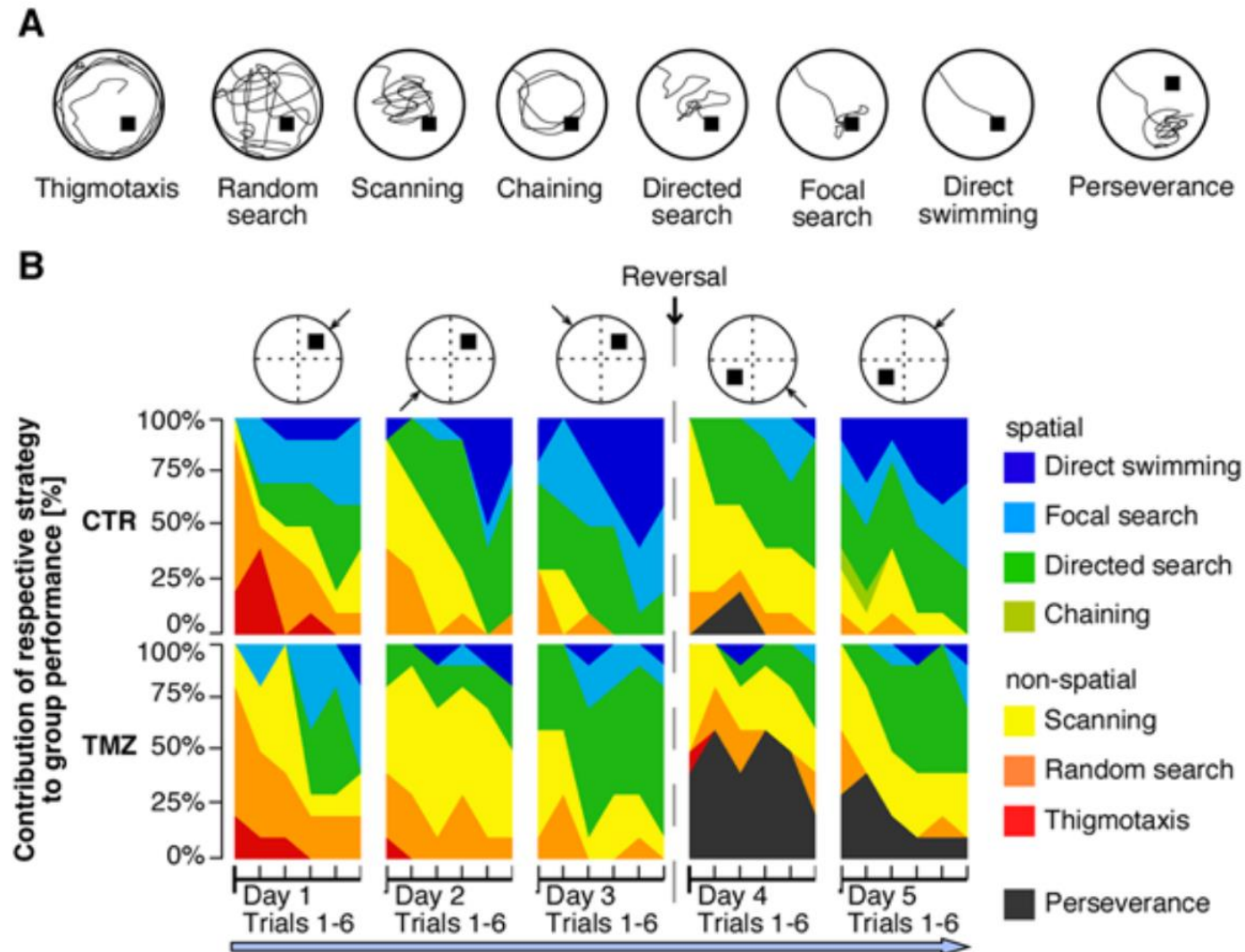
Breathing
pressure sensor



Graboski et al (in prep)

Liao et al. 2023

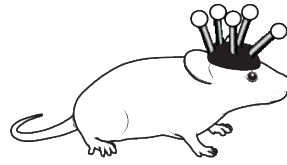
Example 4: Navigation strategies in the space



Summary

- Feature of behavior
 - Hierarchical
 - Spatial and temporal dimension
 - Egocentric and allocentric reference frame
 - Share some common features across animals
- Why analyze behavior
 - Behavior as an output: Ecological meaning
 - Behavior that connect to neural activities
 - Behavior help to explain neural activities
 - Behavior as an window to infer the internal state
- How to analyze behavior
 - Stereotype behavior
 - Behavior is interacting with the environment

Dataset

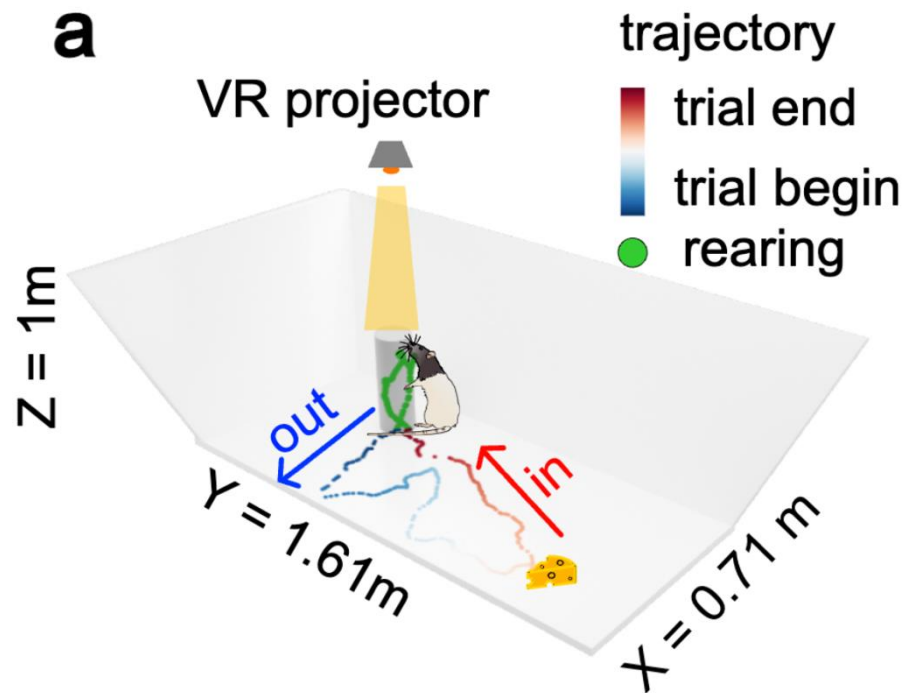


Rat navigation:

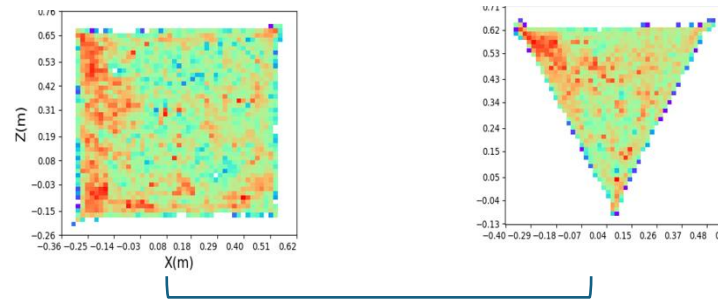
Trial based, 3D trajectory tracking

Mice open field foraging
3D trajectory tracking+ video + head
direction cells

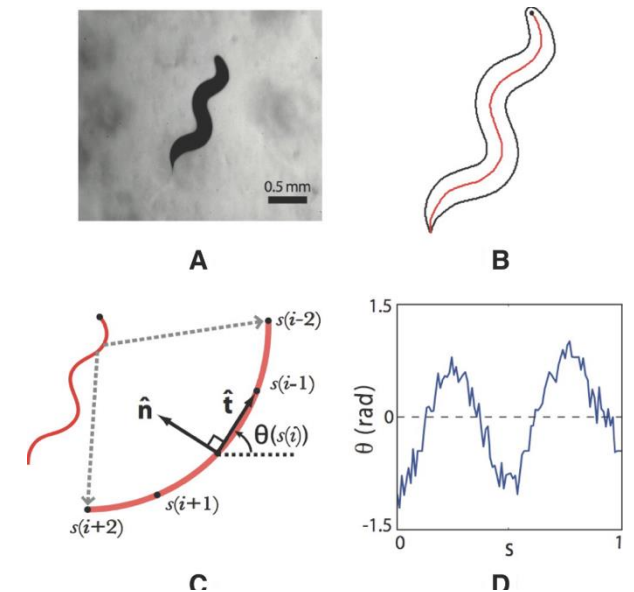
C. Elegance data
Angle of 100 segments



(Fabian Stocek)



(Adrian J Duszkievicz)



A Markovian dynamics for *Caenorhabditis elegans* behavior across scales

Antonio C. Costa^{a,1,2}, Tosif Ahamed^b, David Jordan^c, and Greg J. Stephens^{a,d}

Schedule

- Day1: introduction, fundamental concept for behavior
- Day2: Deeplabcut
- Day3: Segmentation, trajectory analysis
 - Segmentation: Edelhoff 2016 - Path segmentation for beginners
 - Mouse: Findley 2021 - Sniff-synchronized, gradient-guided olfactory search
- Day4: From feature to states
 - C.elegance: Costa 2024 - A Markovian dynamics for C. elegans
 - Moseq: Wiltschko 2015 - Mapping Sub-Second Structure in Mouse Behavior Weinreb 2024 - Keypoint-MoSeq
 - Fruit fly: Berman 2016 - Predictability and hierarchy in Drosophila
- Day5: Project and presentation

Questions?

Day 1: Linear Algebra Foundations for Behavioral Analysis

Python for Animal Behavioral Analysis Workshop

Learning Objectives

By the end of this notebook, you will be able to:

1. **Understand vectors** as both geometric objects (arrows) and algebraic objects (lists of numbers)
2. **Perform vector operations** including addition, dot products, and normalization
3. **Work with coordinate systems** and understand how to change between different frames of reference
4. **Apply rotation matrices** to transform coordinates between reference frames
5. **Compose multiple transformations** using matrix multiplication

Day 1 Rotation & Coordinate Transformations in Animal Tracking

Learning Objectives

By the end of this session, you will be able to:

- Understand 3D rotation matrices and their properties
- Calculate rotation matrices from multi-point animal tracking data
- Convert rotation matrices to meaningful Euler angles (roll, pitch, yaw)
- Understand the difference between intrinsic and extrinsic rotations
- Compare Euler angles vs quaternions for representing orientations
- Transform coordinates between egocentric (animal-centered) and allocentric (world-centered) reference frames